

ERFURT WEIMAR, Germany

WMO: 105540

Lat: 50.9828N

Lon: 10.9608E

Elev: 322

StdP: 97.52

Time Zone: 1.00 (EUC)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.9	-9.0	-14.1	1.2	-10.5	-11.2	1.5	-8.0	14.4	7.6	12.8	6.0	3.2	210	0.579

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.1	30.1	19.2	28.0	18.6	26.0	18.0	20.2	27.2	19.5	26.2	18.6	24.7	3.6	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.1	13.5	22.0	17.1	12.7	21.3	16.2	12.0	20.6	59.4	27.5	56.4	26.1	53.9	24.6	25.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.0	9.5	8.4	DB	-14.4	33.9	3.5	1.5	-16.9	34.9	-18.9	35.8	-20.9	36.6	-23.4	37.6	(4)
(5)			WB	-14.6	22.5	3.5	1.0	-17.1	23.2	-19.1	23.8	-21.1	24.4	-23.6	25.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	9.4	0.0	0.5	4.3	9.5	12.9	16.6	18.9	17.9	14.5	9.8	5.1	2.1	(6)
(7)		DBStd	7.74	5.05	4.84	4.52	4.10	4.05	3.62	3.50	3.37	3.38	3.95	4.04	4.43	(7)
(8)		HDD10.0	1291	311	266	181	59	18	1	0	0	2	53	154	246	(8)
(9)		HDD18.3	3394	569	499	434	267	176	76	37	48	123	264	397	504	(9)
(10)		CDD10.0	1064	0	0	4	42	106	198	274	246	138	47	7	1	(10)
(11)		CDD18.3	127	0	0	0	1	6	22	53	36	9	1	0	0	(11)
(12)		CDH23.3	1168	0	0	0	6	53	207	494	327	77	2	0	0	(12)
(13)		CDH26.7	337	0	0	0	1	8	50	167	99	12	0	0	0	(13)
(14)	Wind	WSAvg	4.0	4.8	4.4	4.4	3.7	3.7	3.5	3.6	3.5	3.7	3.8	4.2	4.9	(14)
(15)	Precipitation	PrecAvg	515	25	24	35	41	62	60	64	59	43	37	39	33	(15)
(16)		PrecMax	764	74	53	96	134	194	115	203	146	120	113	105	78	(16)
(17)		PrecMin	290	0	1	5	8	10	5	9	14	9	3	0	1	(17)
(18)		PrecStd	108	13	14	20	23	34	26	36	33	22	25	22	19	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.0	12.8	18.2	23.9	27.8	31.0	33.8	32.9	28.1	23.0	16.2	12.2	(19)
(20)			MCWB	9.8	7.2	10.7	14.2	18.6	19.8	19.9	19.2	17.6	15.9	13.0	9.4	(20)
(21)		2%	DB	10.0	10.1	15.2	21.1	24.8	28.0	30.9	29.2	25.8	19.2	14.0	10.9	(21)
(22)			MCWB	7.9	7.2	9.7	13.4	17.5	18.9	19.0	19.0	17.0	14.8	11.5	8.7	(22)
(23)		5%	DB	8.2	8.2	13.0	19.1	22.2	25.9	28.2	26.9	23.1	17.2	12.8	9.1	(23)
(24)			MCWB	6.7	6.1	8.8	12.5	15.5	18.4	18.6	18.5	16.3	13.6	10.2	7.2	(24)
(25)		10%	DB	6.9	6.9	10.9	16.9	20.1	23.5	26.0	24.8	20.8	15.8	11.0	7.8	(25)
(26)			MCWB	5.6	5.0	7.7	11.6	14.4	17.3	17.9	17.2	15.5	13.0	8.9	6.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.1	9.0	12.1	15.7	20.0	21.7	21.1	21.1	19.1	16.4	13.2	10.1	(27)
(28)			MCDWB	11.8	10.4	16.4	21.9	25.4	27.9	28.4	27.8	25.8	20.7	15.6	11.9	(28)
(29)		2%	WB	8.2	7.5	10.4	14.1	18.0	19.9	20.0	19.9	17.9	15.2	11.6	8.9	(29)
(30)			MCDWB	9.8	9.6	14.1	19.7	23.2	26.3	27.5	27.1	23.7	19.1	13.8	10.5	(30)
(31)		5%	WB	6.8	6.2	9.3	13.1	16.5	18.6	19.4	19.0	16.8	14.0	10.3	7.4	(31)
(32)			MCDWB	8.1	8.2	12.4	18.2	21.1	24.3	26.4	25.7	22.0	16.9	12.3	9.1	(32)
(33)		10%	WB	5.3	4.8	7.9	12.0	15.1	17.6	18.6	18.1	15.8	13.0	9.2	6.2	(33)
(34)			MCDWB	6.3	6.4	10.8	16.4	19.4	22.6	24.7	23.3	19.7	15.5	10.9	7.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	4.6	5.7	7.6	9.8	9.4	9.8	10.1	10.0	9.0	7.1	5.0	4.6	(35)
(36)			MCDBR	5.5	7.9	11.9	13.4	12.3	13.2	14.0	13.7	12.8	9.7	6.6	6.0	(36)
(37)			MCWBR	4.5	5.7	7.3	7.3	6.7	6.6	5.2	5.7	6.0	5.9	4.9	4.7	(37)
(38)		5% WB	MCDWB	5.3	7.3	10.4	12.0	11.4	12.3	11.7	11.6	10.6	9.0	6.1	5.5	(38)
(39)			MCWBR	4.4	5.6	6.9	6.8	6.5	6.4	4.9	5.4	5.6	5.8	4.8	4.6	(39)
(40)			Clear-Sky Solar Irradiance	taub	0.303	0.326	0.364	0.403	0.407	0.404	0.419	0.412	0.377	0.350	0.327	0.299
(41)	taud	2.408		2.368	2.293	2.224	2.243	2.272	2.235	2.277	2.358	2.425	2.428	2.401	(41)	
(42)	Ebn at Noon	725		801	829	834	848	853	831	817	808	762	686	668	(42)	
(43)	Edh at Noon	61		82	106	127	130	128	131	119	99	77	60	53	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.75	1.46	2.54	4.04	4.87	5.29	4.99	4.29	3.09	1.80	0.84	0.57	(44)	
(45)		RadStd	0.12	0.21	0.32	0.51	0.57	0.45	0.63	0.43	0.42	0.21	0.12	0.07	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (48 neighbors)	N/A	N/A	N/A	N/A	N/A	+0.37	N/A	N/A	N/A	N/A

Nomenclature: See separate page